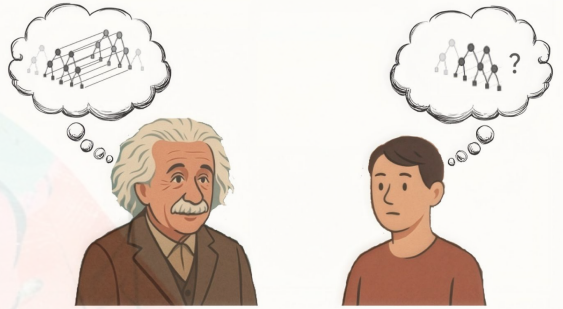
Measuring Combinatorial Creativity in Large Language Models and Humans





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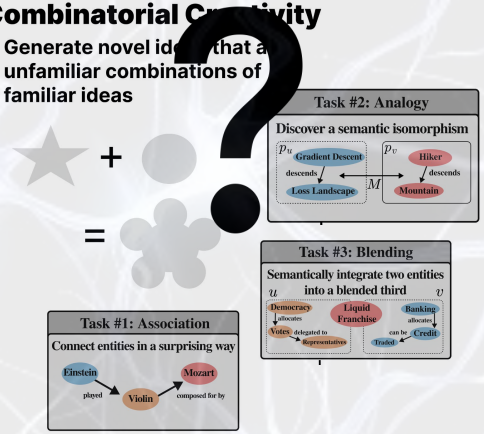
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Types of Creative Processes & Products

Boden's types

Combinatorial Creativity

Generate novel ideas that are unfamiliar combinations of familiar ideas



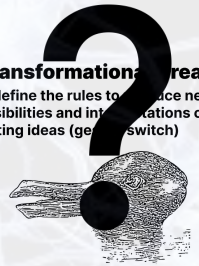
Exploratory Creativity

Generate novel ideas by following existing rules



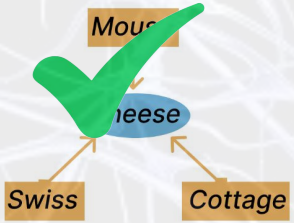
Transformational Creativity

Re-define the rules to produce new possibilities and interpretations of existing ideas (generative switch)



Divergent vs. convergent thinking

Remote Associates Test



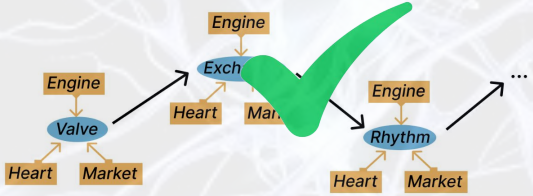
Convergent thinking

Divergent Association Task



Divergent thinking

Divergent Remote Association Test



Convergent + Divergent thinking

Abduction



(a) Geocentric conceptual space S (b) Heliocentric conceptual space S'

Figure 1: Conceptual-space DAGs before (S) and after (S') the Co… and Keplerian revolution. All non-axiom artifacts reach at least one axiom (sink node), satisfying the formal definitio… conceptual space.

Table 1: Historical instances of transformational creativity in science, demonstrating how the transformation of axioms $A \to A'$ transformed the conceptual spaces $S \to S'$.

S (Initial Space)	A (Selected Axioms)	A' (Mo… …oms)	S' (Transformed Space)
Newtonian Physics	Absolute time; separate space/time; gravity as force	Relative time; spacetime unity; gravity as curvature	Relativistic Physics
Ptolemaic Astronomy	Earth-centered universe	Sun-centered model	Heliocentric system
Euclidean Geometry	Parallel Postulate: through a point not on a line there is exactly one parallel line	Altered Parallel Postulate: through such a point there are *zero* (elliptic) or *at least two* (hyperbolic) parallels	Non-Euclidean geometry (elliptic and hyperbolic)

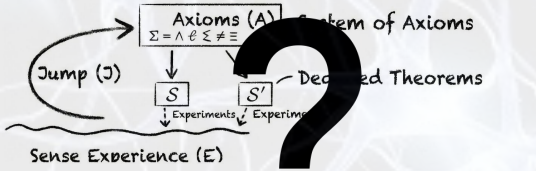


Figure 1. A generative AI reconstruction of Einstein's E-J-A diagram. Einstein drew this diagram i… …ter to Maurice Solovine, showing a cyclical line jumping fro… …se Experience (E) to Axioms (A) via a Jump (J), and then deducing logical consequences. Ironically, the hallucination of the axiomatic symbols highlights the very difficulty of automating the jump.

Combinatorial Creativity

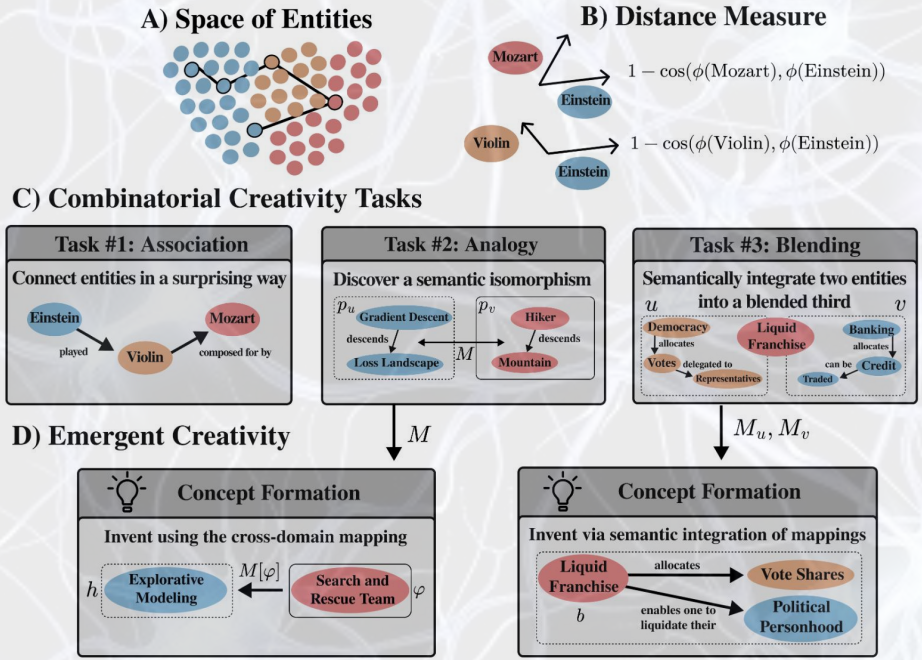
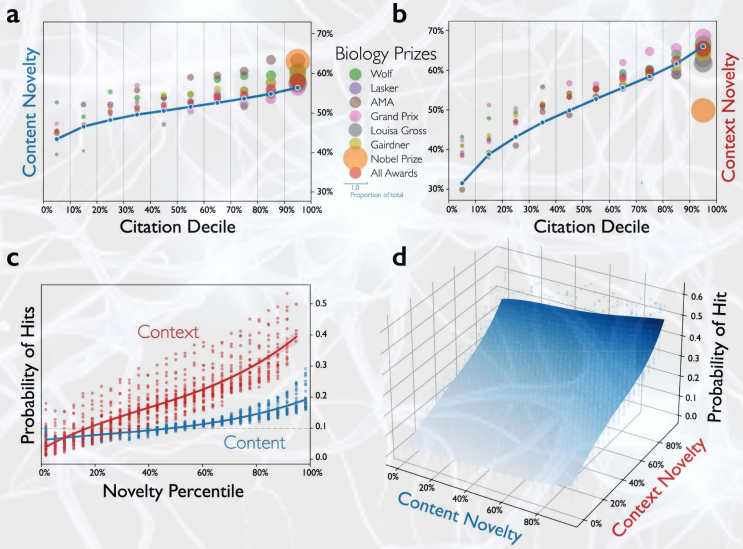


Figure 2: **An overview of the KOMBINE benchmark.** **(A)** Combinatorial creativity involves combining entities in original, useful, and surprising ways. **(B)** The semantic distance between entities is commonly used to calculate surprise and originality. **(C)** Combinatorial creativity largely falls into one of three types: (1) *Association*, (2) *Analogy*, and (3) *Blending*. **(D)** Analogies and blends support original concept invention via structure mapping and semantic integration.

Table 1: **Forms of combinatorial creativity assessed in KOMBINE.** Each task assesses an ability recognized in cognitive science as integral to combinatorial creativity (Boden, 2004; Simonton, 2023)

Task (§3)	Creativity Mechanism(s) & Source(s)	Example
Association	Remote association (Mednick, 1962)	*Einstein* → *violin* → *Mozart*
Analogy	Analogy (Gentner, 1983; Rumelhart and Abrahamson, 1973) and metaphor (Lakoff and Johnson, 2008)	*immune system* : *body* :: *national defense* : *nation*
Blending	Conceptual blending (Fauconnier and Turner, 2008); conceptual combination (Thagard, 1984)	*organism* + *machine* → *cyborg*



Combinatorial Creativity Tasks

Table 2: **Example outputs on each task**. Colors indicate the domain of each input (u, v), the **invented entity**, and **emergent structure** of the blend. Images are for illustration purposes only.

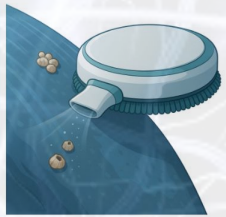
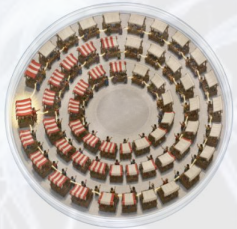
Task (model)	Example output	Invention illustration
(a) Association (claude-opus-5)	**Inputs** (u, v): *Radio* → *The Milky Way*. **Path** (p): (Radio, aired hit, Vincent (song)) → (Vincent (song), tribute to, Vincent van Gogh) → (Vincent van Gogh, painted, The Starry Night) → (The Starry Night, swirls depict, The Milky Way).	N/A
(b) Analogy (claude-fable-5)	**Inputs** (u, v): *The blue whale* :: *The mattress*. **Path a** (p_u): (The blue whale, covered by, epidermis), (epidermis, hosts, barnacles), (barnacles, feed on, plankton). **Path b** (p_v): (The mattress, covered by, ticking), (ticking, hosts, dust mites), (dust mites, feed on, skin flakes). **Source** (φ): vacuum cleaner (from The mattress, no counterpart in The blue whale). **Invention** $(h \leftarrow M[\varphi])$: **whale groomer drone**. **Source to Target Mapping** $(M^{u,v})$: (vacuum cleaner, removes, dust mites) ↦ (whale groomer drone, removes, barnacles) (vacuum cleaner, cleans, ticking) ↦ (whale groomer drone, cleans, epidermis) (vacuum cleaner, maintains, The mattress) ↦ (whale groomer drone, maintains, The blue whale)	
(c) Blending (claude-opus-4.5)	**Inputs** (u, v): *Adam Smith* + *Bacteria*. **Generic Space** (g): autonomous agents whose local signals generate collective order without central control. **Invention** (b): **market colony**. **Blended Space** (c'), each row tagged by its projection: (market colony, coordinates through, local signals) [uv] (market colony, produces, emergent efficiency) [uv] (market colony, specializes via, division of labor) [u] (market colony, adapts through, replication) [v] (market colony, evolves its, exchange protocols) [*emergent*] (market colony, spreads, practices contagiously) [*emergent*]	

Table 5: **The entity pool $\mathcal{C}$.** All 283 entities, categorized by domain.

Domain	Entities
science	Gravity, Evolution, Entropy, Photosynthesis, Electricity, Radioactivity, The atom, Quantum mechanics, The theory of relativity, Plate tectonics, The Big Bang, Natural selection, The periodic table, Nuclear fission, Magnetism, The speed of light
mathematics	Zero, Infinity, Pi, Prime numbers, The Pythagorean theorem, Calculus, Probability, Geometry, The golden ratio, Fractals, Chaos theory, Set theory, Logarithms, Fibonacci sequence
technology	The printing press, The steam engine, The internet, The telephone, The computer, The transistor, The automobile, The airplane, The light bulb, The wheel, Artificial intelligence, The battery, Radio, GPS, The camera, The clock
biology	DNA, The cell, The brain, The heart, Bacteria, Viruses, The immune system, Photosynthesis, The human genome, Coral reef, The honey bee, The lion, The octopus, The oak tree, Mushrooms, The blue whale
medicine	Penicillin, Vaccines, Anesthesia, The stethoscope, Antibiotics, Surgery, X-rays, Insulin, The Hippocratic Oath, Blood transfusion, Aspirin, The microscope
art	The Mona Lisa, Impressionism, Cubism, Leonardo da Vinci, Pablo Picasso, Vincent van Gogh, Frida Kahlo, Michelangelo, The Sistine Chapel, Surrealism, Sculpture, Andy Warhol, Calligraphy, Origami
music	Jazz, The symphony, The piano, Ludwig van Beethoven, The Beatles, Wolfgang Amadeus Mozart, Hip hop, The guitar, Opera, Bob Dylan, The blues, Reggae, The orchestra, Rock and roll
literature	William Shakespeare, The novel, Poetry, Homer's Odyssey, Don Quixote, Jane Austen, Greek tragedy, Haiku, Leo Tolstoy, Franz Kafka, The sonnet, Mythology, Science fiction, Fairy tales
film	Cinema, Alfred Hitchcock, Charlie Chaplin, Animation, Akira Kurosawa, Documentary film, Steven Spielberg, Film noir, Walt Disney, The Western
philosophy	Democracy, Free will, Justice, Aristotle, Plato, Socrates, Existentialism, Ethics, Immanuel Kant, Stoicism, Confucius, The social contract, Logic, Nihilism
religion	Buddhism, Christianity, Islam, Hinduism, Judaism, Meditation, The Ten Commandments, Karma, Pilgrimage, Zen, Prayer, Mythology
history	The Roman Empire, The Renaissance, The French Revolution, The Industrial Revolution, Ancient Egypt, The Silk Road, World War II, The Cold War, The Enlightenment, Feudalism, The Great Depression, Colonialism, The Space Race
geography	Mount Everest, The Amazon rainforest, The Sahara, The Nile, The Great Barrier Reef, Antarctica, The Grand Canyon, The Himalayas, Volcanoes, The Pacific Ocean, Iceland, Japan, The Alps, Niagara Falls
politics	Abraham Lincoln, Nelson Mandela, Mahatma Gandhi, Napoleon, Winston Churchill, The United Nations, Cleopatra, Julius Caesar, The Constitution, Martin Luther King Jr., Diplomacy, The monarchy
economics	Money, The stock market, Capitalism, Supply and demand, Adam Smith, Inflation, Banking, Trade, Cryptocurrency, The Great Depression, Taxation, Globalization
sports	Chess, Football (soccer), The Olympic Games, Boxing, Muhammad Ali, Basketball, Marathon running, Tennis, Gymnastics, Surfing, Cricket, Formula One
food	Coffee, Chocolate, Wine, Bread, Rice, Tea, Cheese, Sushi, Pizza, Chili peppers, Salt, Honey, Fermentation, Spices
everyday	The mirror, The lock and key, Glass, Paper, The umbrella, Scissors, Soap, The mattress, Candles, The map, The button, Rope, The needle, The compass
nature	Fire, Lightning, The rainbow, Snow, The tides, Earthquakes, Diamonds, Gold, Rivers, Deserts, The seasons, Fog, Amber, Crystals
cosmos	The Sun, The Moon, Mars, The solar system, Black holes, The Milky Way, Comets, The Northern Lights, Saturn's rings, Eclipses, Meteorites, Constellations
ideas	Time, Language, Color, Symmetry, Chaos, Memory, Consciousness, Beauty, Randomness, Emergence, Feedback loops, Metaphor, Paradox, Networks

Evaluating Creativity (analogy example)

(b) Analogy (claude-fable-5)

Inputs (u, v): *The blue whale* :: *The mattress*.
Path a (p_u): (The blue whale, covered by, epidermis), (epidermis, hosts, barnacles), (barnacles, feed on, plankton).
Path b (p_v): (The mattress, covered by, ticking), (ticking, hosts, dust mites), (dust mites, feed on, skin flakes).
Source (φ): vacuum cleaner (from The mattress, no counterpart in The blue whale).
Invention $(h \leftarrow M[\varphi])$: **whale groomer drone**.
Source to Target Mapping $(M^{u,v})$:
(vacuum cleaner, removes, dust mites) $\mapsto$ (whale groomer drone, removes, barnacles)
(vacuum cleaner, cleans, ticking) $\mapsto$ (whale groomer drone, cleans, epidermis)
(vacuum cleaner, maintains, The mattress) $\mapsto$ (whale groomer drone, maintains, The blue whale)



Utility {0,1}

Is it a valid structure mapping?

Each path (p_u, p_v) must be **factual**, and the two paths must use **identical relations**.

Surprise [0, 1]

How distant are the added concepts used in the analogical mapping?

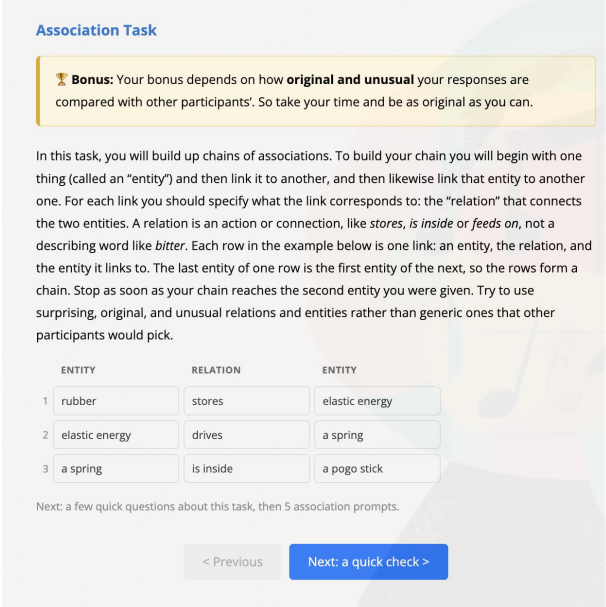
d(**epidermis**, **ticking**)
d(**barnacles**, **dust mites**)
d(**plankton**, **skin flakes**)

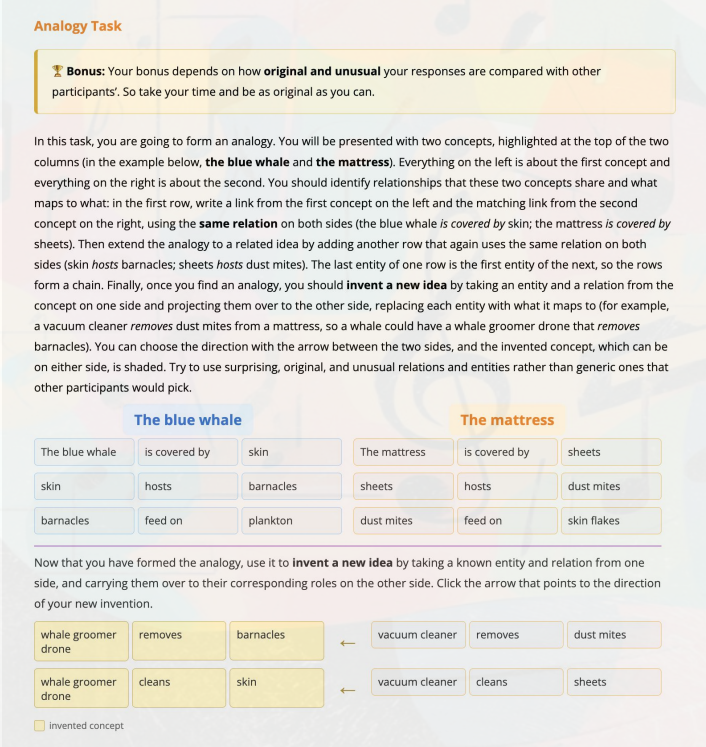
Originality [0, 1]

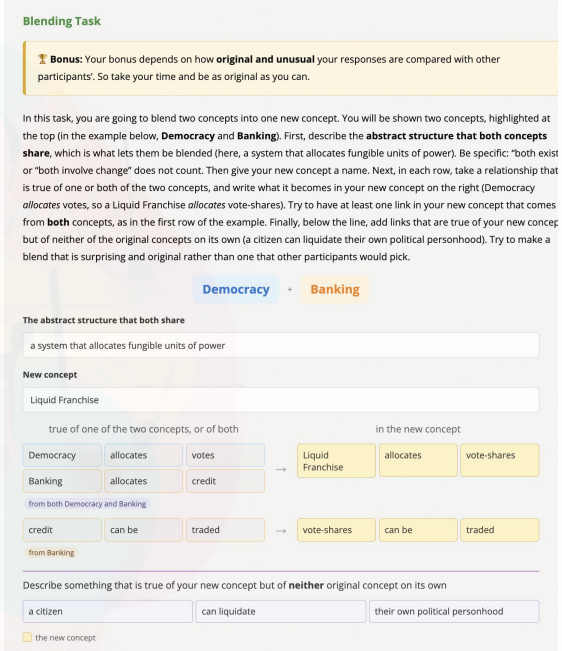
How distant are the concepts, relations used in this response relative to the set of those from every other participant?

d(**epidermis**, {skin, skin grooves})
d(**ticking**, {fabric, seams})
d(**barnacles**, {barnacle, skin})
d(**dust mites**, {dust, house dust})
...

Human Study Platform







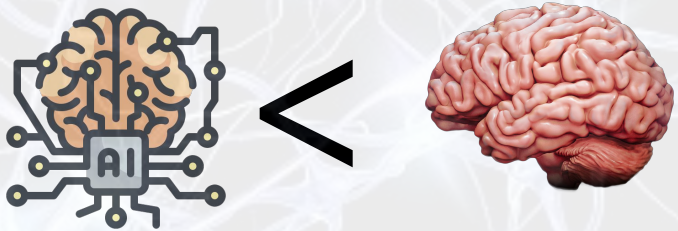
75 human participants (Statistics: 32 M, 40 F, 1 non-binary, 1 genderfluid, 1 preferred not to say; Mean age = 40.1 years, SD = 12.7, range=[20, 78])

Who is more combinatorially creative?

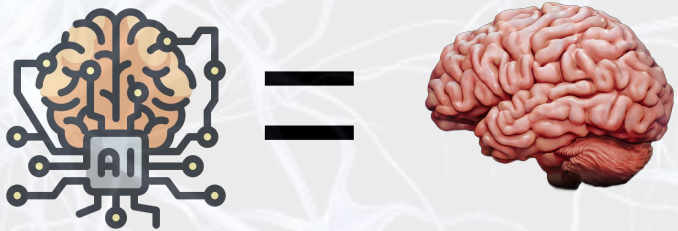
(a) LLMs



(b) The average human



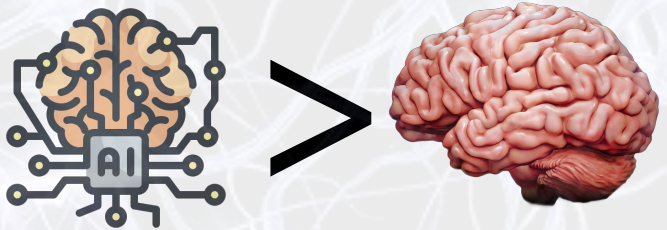
(c) About equal



LLMs are more combinatorially creative than the average human

Table 3: **Top systems by combinatorial creativity.** The highest-scoring systems on *Overall* and on each task, as the equal-weight composite of each scoring dimension in Section 3.

Rank	Overall	Association	Analogy	Blending
1	gpt-5.6-sol (55.2)	claude-opus-4.6 (51.4)	claude-opus-4.6 (63.1)	gpt-5.6-sol (64.8)
2	gpt-6-astra (54.4)	gpt-5.2 (50.3)	gpt-6-astra (61.1)	gemini-3.1-pro (59.5)
3	claude-opus-4.6 (52.0)	gemini-2.5-flash (48.3)	grok-4.6 (60.1)	gpt-6-astra (56.5)
4	gemini-3.1-pro (48.7)	claude-opus-4.5 (48.0)	gpt-5 (58.6)	gemini-3.7-flash (52.9)
5	claude-fable-5 (48.1)	gpt-5-mini (47.9)	grok-4.5 (55.4)	claude-fable-5 (45.7)
6	grok-4.6 (47.6)	grok-4.6 (47.8)	gpt-5.6-sol (55.4)	claude-fable-5.1 (43.3)
7	gpt-5 (47.4)	grok-4.5 (47.5)	claude-sonnet-4.6 (54.3)	claude-opus-4.6 (41.4)
8	grok-4.5 (46.9)	claude-fable-5 (47.3)	claude-fable-5 (51.3)	gpt-5.2 (39.1)
35	phi-4 (17.4)	glm-4.5-air (19.6)	***Humans*** (19.1)	***Humans*** (7.4)
36	***Humans*** (13.7)	***Humans*** (14.7)	kimi-k2 (17.0)	phi-4 (4.4)



Are rankings robust to embedding model?

	Overall	Per-task		
Encoder pair	ρ	Assoc.	Analogy	Blend
MiniLM ∼ BGE	0.999	0.996	0.997	0.999
MiniLM ∼ E5	0.998	0.996	0.996	0.997
BGE ∼ E5	0.998	0.996	0.999	0.999

Table 11: **Leaderboard agreement across embedding models.** Spearman ρ between the overall rankings of the 35 models under each pair of encoders, as well as the Spearman ρ by task.

Dimension	ICC(2,3)	Cicchetti (1994)
Blend semantic integration quality ($Q(c')$)	0.67	good
Blend utility ($G(g)$)	0.67	good
Analogy integration quality ($Q(h)$)	0.56	fair
Analogy utility ($V(h)$)	0.55	fair
Blend utility ($V(c')$)	0.48	fair

Table 6: **Inter-judge reliability of the three judge panel**, as assessed by the intra-class correlation coefficient (Cicchetti, 1994). ICC(2,3) is the reliability of the mean of the three judges.

Do LLM judges agree?

However, LLMs tend to homogenize

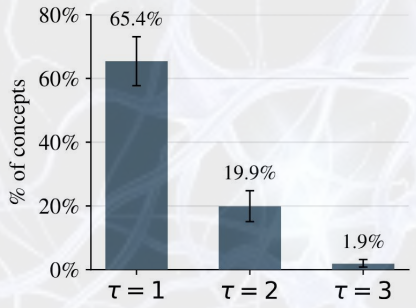
Table 4: **Examples of two inventive multiples.** Semantically indistinguishable properties are placed side-by-side and highlighted. *Operators*: Blending (+), Analogy (::).

Opera + Documentary film

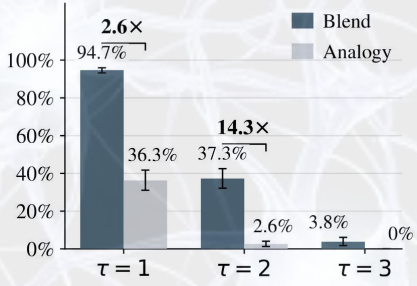
claude-fable-5.1 **"testimony opera"**	cos	gemini-3.7-flash **"docu-opera"**
p_2: intercuts archival footage	0.87	p'_2: incorporates archival footage
p_4: derives melody from recorded speech	0.81	p'_4: derives melody from spoken pitch contours
p_3: halts action for sung testimony	0.77	p'_3: delivers sung verbatim testimony
p_1: accompanied by orchestra		p'_1: utilizes orchestral overtures

Democracy :: Banking

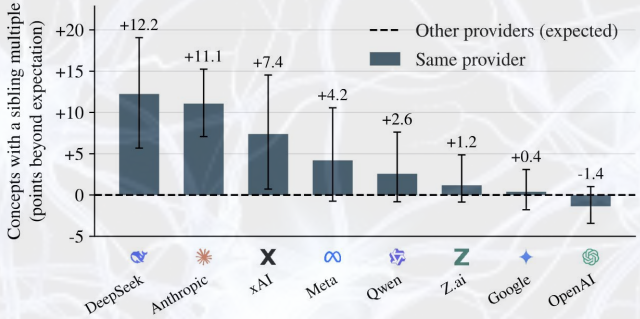
claude-fable-5 **"vote insurance"**	cos	claude-fable-5.1 **"legitimacy stress test"**
p_1: protects ballot	0.84	p'_3: protects votes
p_5: mandated by election commission	0.70	p'_4: is overseen by electoral commission
p_2: restores popular will		p'_1: simulates recall election
p_3: triggered by election annulment		p'_2: measures mandate
p_4: prevents voter boycott		



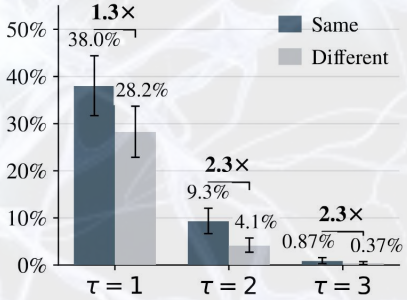
(a) Concepts in a multiple



(b) By task

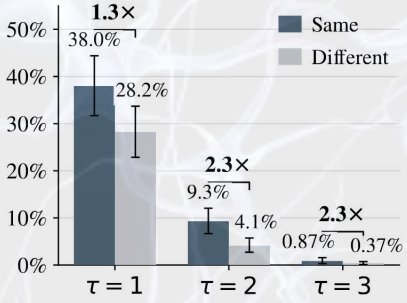


(d) By provider



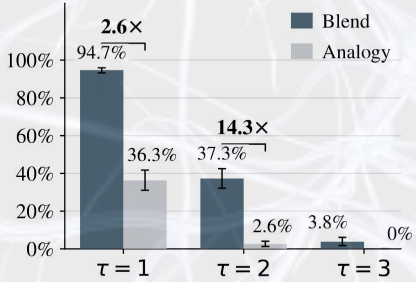
(c) By family

Models from the same provider are 1.3-2.3x more likely to invent multiples

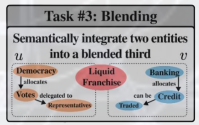


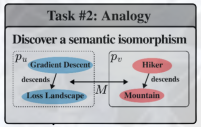
(c) By family

Models are 2.6-14x more likely to invent multiples on the blending vs. analogy task

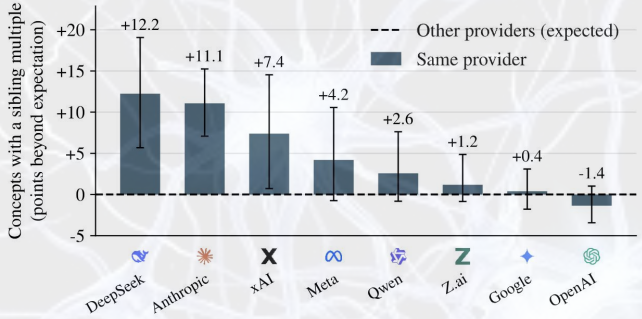


(b) By task





DeepSeek and Anthropic models most likely to produce multiples; OpenAI and Google models least likely



(d) By provider

And LLMs exhibit an *Abstraction Bottleneck*

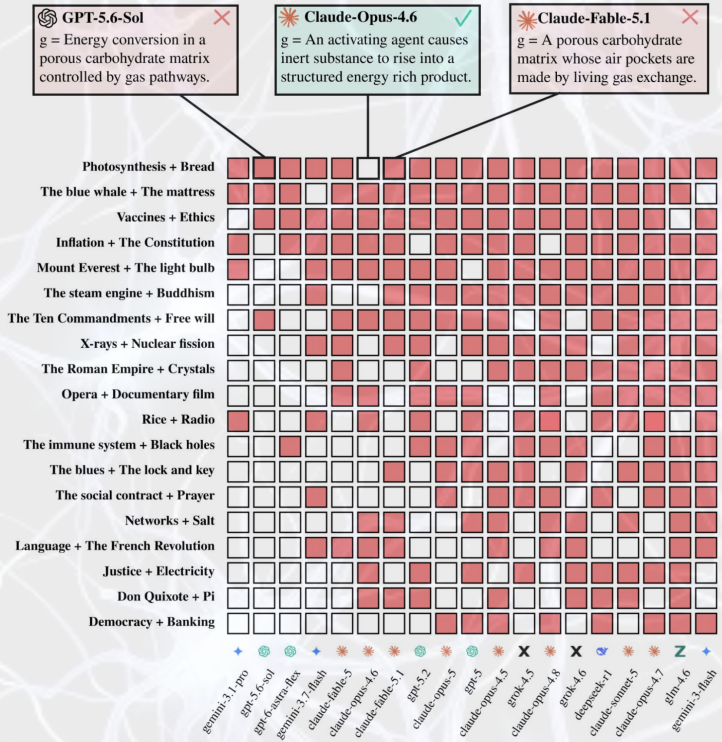


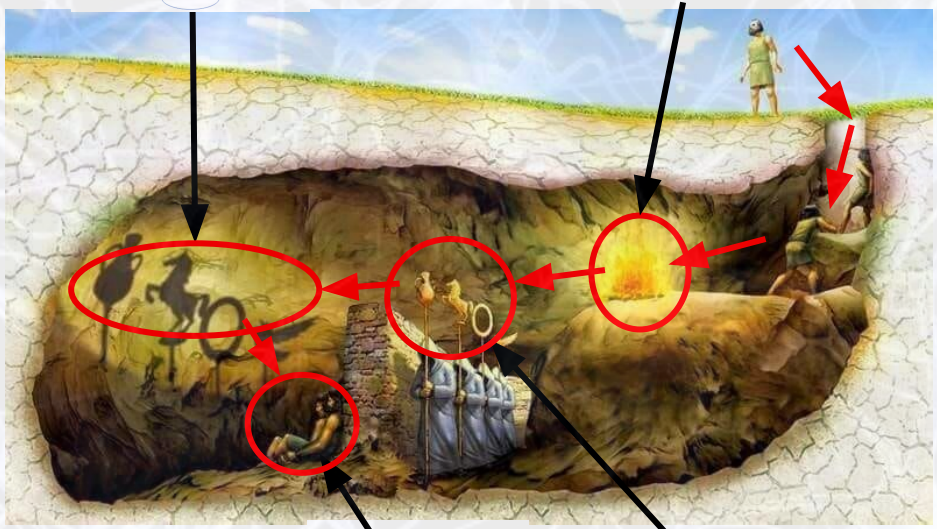
Figure 4: **Blending generic space failures among frontier models.** A red cell marks an item on which a model's generic space was invalid, often due to applying only to one input. Although blending is difficult for LLMs, no row is fully red, i.e., every entity pair had *at least one model* obtain a valid generic space.



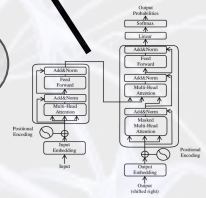
Training text 3



Human experience 1

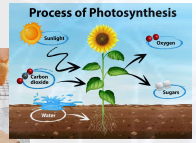


4



LLMs

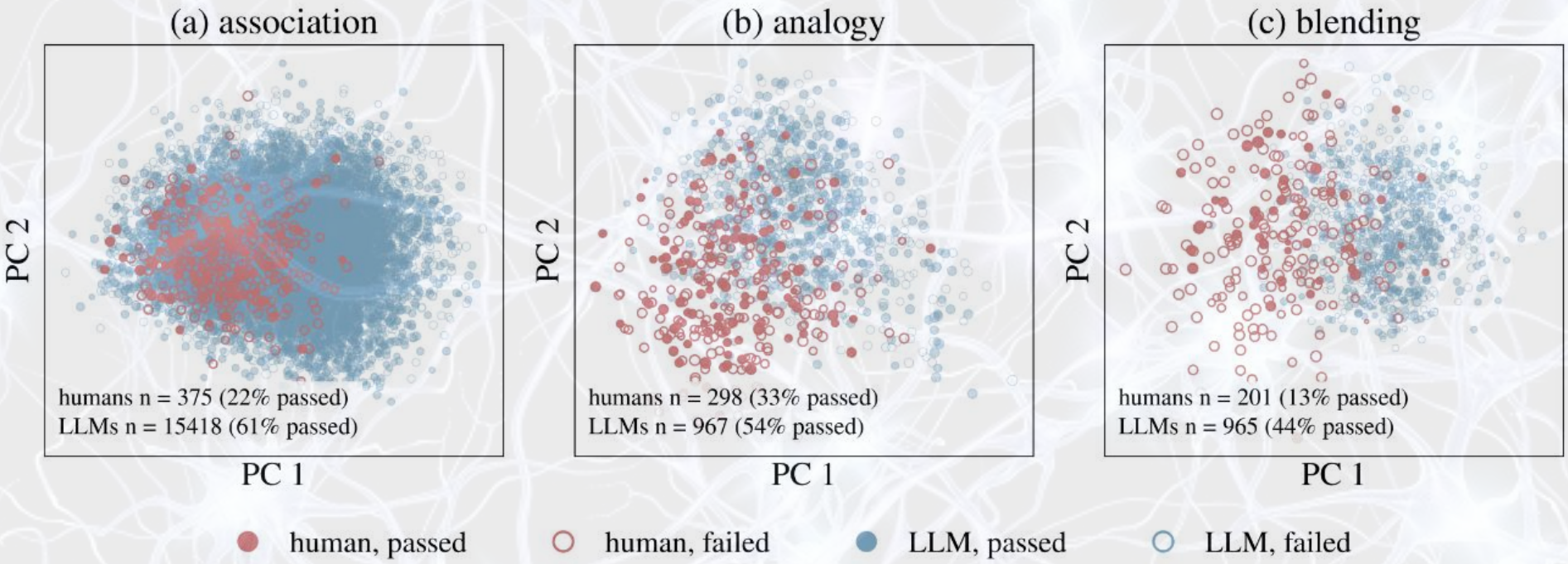




2

Entities

Humans, LLMs Combine in Distinct Regions of Semantic Space



Humans, LLMs Combine in Distinct Regions of Semantic Space

Human-generated **concepts** (entities) more likely to be: **everyday objects**, **states**, and **feelings** (e.g., “rock,” “happiness,” “love”)



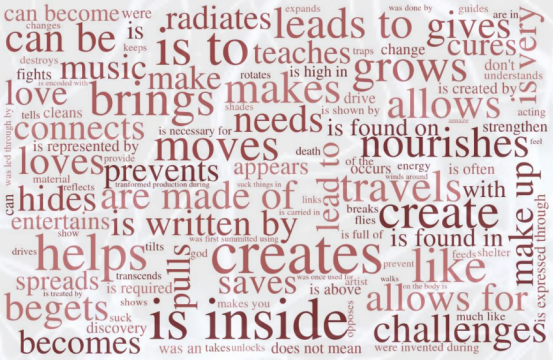
(a) Human concepts



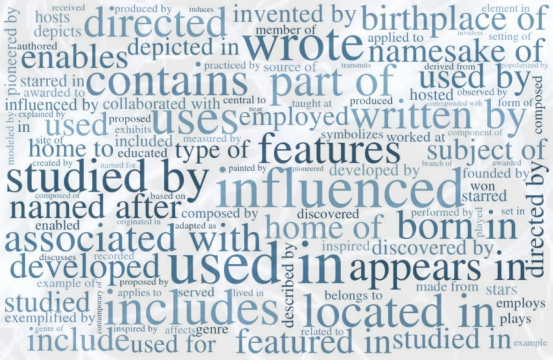
(b) LLM concepts

LLM-generated **concepts** more likely to: name **people** and **places** (e.g., “James Clerk Maxwell,” “Galileo Galilei,” “London”)

Human-generated **relations** more likely to express **motion** and **plain states of being** (e.g., “leads to,” “moves,” “is inside”)



(c) Human relations



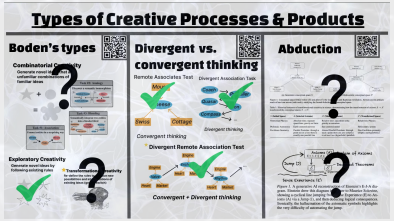
(d) LLM relations

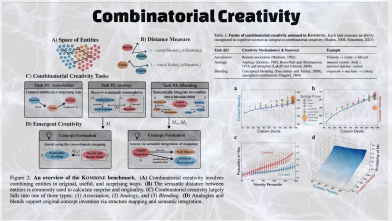
LLM-generated **relations** more likely to express **creation**, **communication**, and **cognition** (e.g., “wrote,” “named after,” “studied by”)

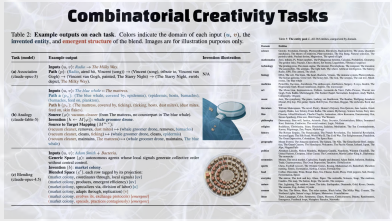
The Future of Human-AI Co-Creativity

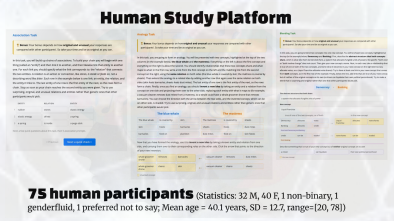
Open Question #1: Models show greater combinatorial breadth, but do they outperform human domain experts in combinatorial depth?

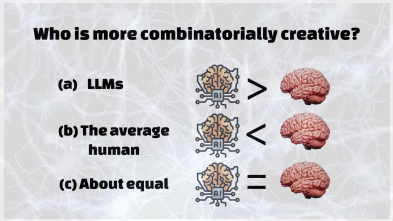
Open Question #2: How should we structure human-AI co-creative interactions to best support, e.g., scientific creativity?

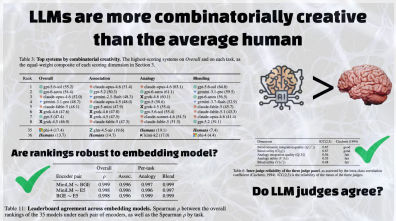


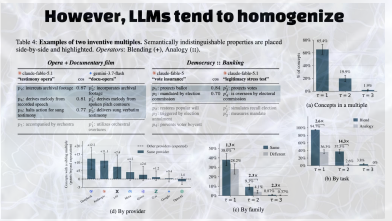


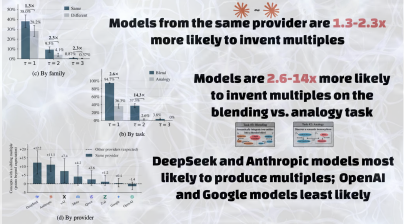


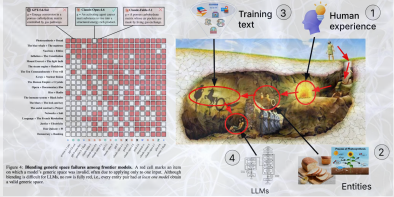


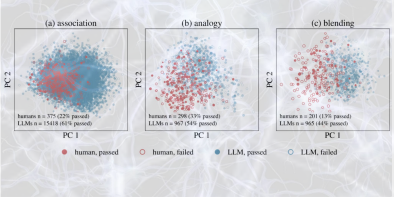
















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Paper: https://arxiv.org/abs/2509.21043
Slides: schapiro.ai/files/kombine-slides

MEASURING COMBINATORIAL CREATIVITY IN LARGE LANGUAGE MODELS AND HUMANS

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Qiawen Ella Liu[8] **Sumuk Shashidhar**[6] **Jonah Black**[6] **Royce Moon**[3, 9]
Dilek Hakkani-Tur[6] **Heng Ji**[6] **Claire E. Stevenson**[7] **Lav R. Varshney**[3]
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Figure 1: **Overview.** Combinatorial creativity, spanning association, analogy, and blending, describes the process of combining familiar ideas in unfamiliar ways, and is recognized as central to scientific and technological progress (Thagard, 2010; Boden, 2004; Shi and Evans, 2023; Simonton, 2021).